

What the Kuramoto Model Got Wrong and How Field-Based Coupling Fixes It

A Structural Resolution of the $N = 2$ Synchronization Failure

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Abstract

The Kuramoto model is the standard framework for analyzing phase synchronization in coupled oscillator systems. While it succeeds in the thermodynamic limit, it exhibits a structural failure at low cardinality. In particular, for $N = 2$ oscillators, the model collapses into a purely antagonistic phase-difference dynamic with no shared convergence structure. This paper identifies the source of that failure and presents a minimal corrective reformulation based on field-driven coupling. By elevating the global phase already implicit in the Kuramoto order parameter into an explicit dynamical attractor, synchronization becomes well-defined for all $N \geq 2$. The resulting formulation preserves Kuramoto behavior at large N while resolving the $N = 2$ degeneracy without introducing additional degrees of freedom.

1. Introduction

The Kuramoto model has become foundational across physics, neuroscience, engineering, and multi-agent systems. Its appeal lies in its simplicity and its success in describing emergent synchronization in large ensembles of oscillators. However, this success relies implicitly on statistical averaging. When the system size is reduced to its minimal nontrivial case, the underlying assumptions break.

This paper focuses on the $N = 2$ case, where the Kuramoto model ceases to represent synchronization as a convergent process and instead reduces to a frequency tug-of-war. This is not a numerical pathology but a structural one. The failure exposes a deeper modeling assumption: synchronization is treated as pairwise interaction rather than alignment to a shared field.

2. The Kuramoto Model and the $N = 2$ Degeneracy

The standard Kuramoto phase evolution equation is

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i).$$

For large N , the system exhibits collective behavior mediated by the emergent order parameter. However, when $N = 2$, the equations reduce to

$$\begin{aligned}\frac{d\theta_1}{dt} &= \omega_1 + \frac{K}{2} \sin(\theta_2 - \theta_1), \\ \frac{d\theta_2}{dt} &= \omega_2 + \frac{K}{2} \sin(\theta_1 - \theta_2).\end{aligned}$$

Defining the phase difference $\Delta = \theta_2 - \theta_1$, subtraction yields

$$\frac{d\Delta}{dt} = \omega_2 - \omega_1 - K \sin(\Delta).$$

This formulation contains no shared reference phase. The dynamics are strictly adversarial, governed solely by frequency mismatch and local coupling. Synchronization is no longer an emergent property but a special-case balance.

3. Missing Structure: The Shared Phase Field

In real synchronizing systems, oscillators do not coordinate by pairwise negotiation alone. They entrain to a collective rhythm that exists at the system level. In the Kuramoto framework, this collective object already exists implicitly in the complex order parameter

$$re^{i\Phi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}.$$

While the magnitude r is used extensively as a synchronization metric, the phase Φ is typically not treated as a governing field. This omission is inconsequential at large N but fatal at small N .

4. Field-Based Coupling Reformulation

The corrective step is minimal. Each oscillator is coupled not directly to other oscillators, but to the shared phase field Φ :

$$\frac{d\theta_i}{dt} = \omega_i + K \sin(\Phi - \theta_i).$$

This reformulation introduces no new state variables. The field Φ is entirely determined by the existing phases. What changes is the topology of interaction: alignment is global rather than pairwise.

5. Resolution of the $N = 2$ Case

For $N = 2$,

$$\Phi = \arg(e^{i\theta_1} + e^{i\theta_2}).$$

Using standard trigonometric identities,

$$e^{i\theta_1} + e^{i\theta_2} = 2 \cos\left(\frac{\theta_2 - \theta_1}{2}\right) e^{i(\theta_1 + \theta_2)/2}.$$

Thus,

$$\Phi = \frac{\theta_1 + \theta_2}{2}.$$

The shared field becomes the phase midpoint. Substituting into the field-coupled equations yields

$$\begin{aligned} \frac{d\theta_1}{dt} &= \omega_1 + K \sin\left(\frac{\theta_2 - \theta_1}{2}\right), \\ \frac{d\theta_2}{dt} &= \omega_2 + K \sin\left(\frac{\theta_1 - \theta_2}{2}\right). \end{aligned}$$

Defining $\Delta = \theta_2 - \theta_1$, the phase difference evolves as

$$\frac{d\Delta}{dt} = \omega_2 - \omega_1 - 2K \sin\left(\frac{\Delta}{2}\right).$$

Stable fixed points exist whenever

$$|\omega_2 - \omega_1| < 2K.$$

Unlike the original Kuramoto reduction, both oscillators now align toward the same attractor. Synchronization emerges through shared-field entrainment rather than antagonistic balance.

6. Structural Interpretation

The failure of the Kuramoto model at $N = 2$ is not accidental. It reflects a limitation of Λ^1 interaction models, where coherence is expected to arise from local coupling alone. Field-based coupling corresponds to a Λ^2 structure, where agents align within a shared harmonic environment.

At large N , Λ^1 and Λ^2 formulations become approximately equivalent. At small N , the distinction is decisive.

7. Conclusion

The Kuramoto model remains a powerful description of collective synchronization, but only within its domain of validity. By explicitly modeling the shared phase field already latent in the system, synchronization becomes structurally well-defined for all system sizes. This correction resolves the $N = 2$ degeneracy without altering large- N behavior and aligns the mathematical model with observed synchronization mechanisms in physical, biological, and engineered systems.

This reformulation completes, rather than replaces, the Kuramoto framework.

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